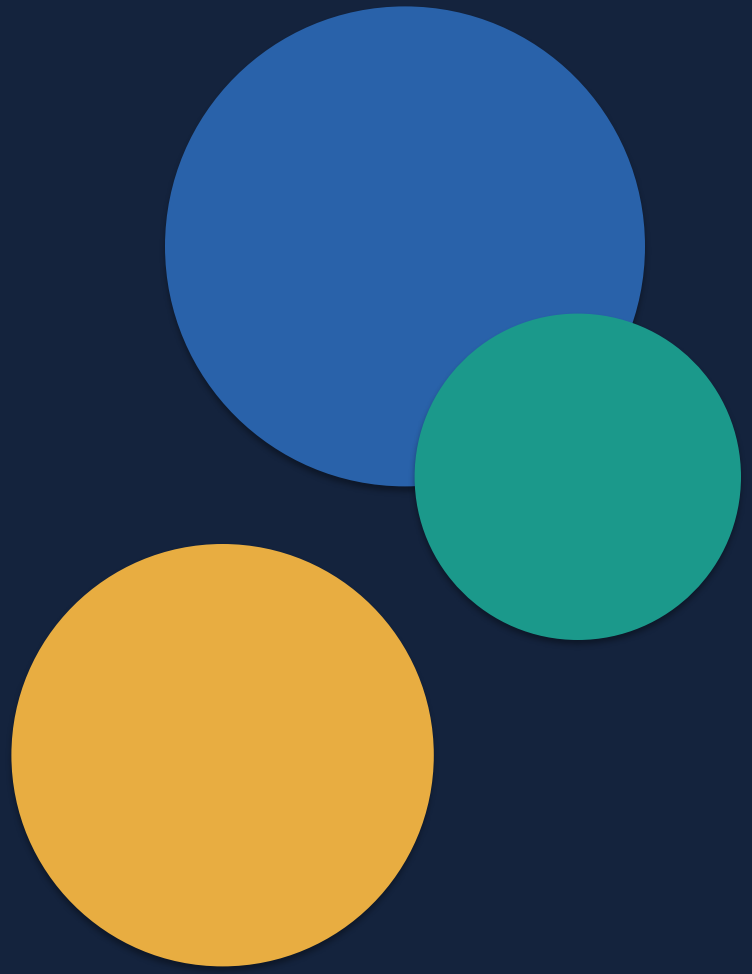
COURSE REGISTRATION • PREDICTIVE ANALYTICS

# Smart Course Registration System

## with Predictive Analytics



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# Project Objective

## The Problem

- Students register blindly without knowing course difficulty
- No real-time credit limit check leads to overload
- Pass-out students (>=140 credits) still occupy seats
- Instructors get enrollment lists days after registration ends
- student may want to drop a course after registration

## Our Solution

- ML risk tags (Low / Medium / High) shown before enrollment
- 40-credit cap enforced automatically via SQL validation
- Graduated students blocked at login itself

Live instructor roster with student names and emails

- students can drop their registered course

## Three User Roles in One Portal

- **Student:** View profile (CGPA, credits), browse department courses with risk badges, add/drop courses with credit validation
- **Instructor:** View assigned sections for Fall 2026, see room number, schedule, and enrolled student roster (Name, ID, Email)
- **Admin:** View list of all registered academic departments

# System Architecture

### 1 User interface

HTML templates

- Login
- Student
- Instructor
- Admin

### 2 Flask application

Routes and session checks

Business rules and SQL queries

Render templates with database results

### 3 MySQL database

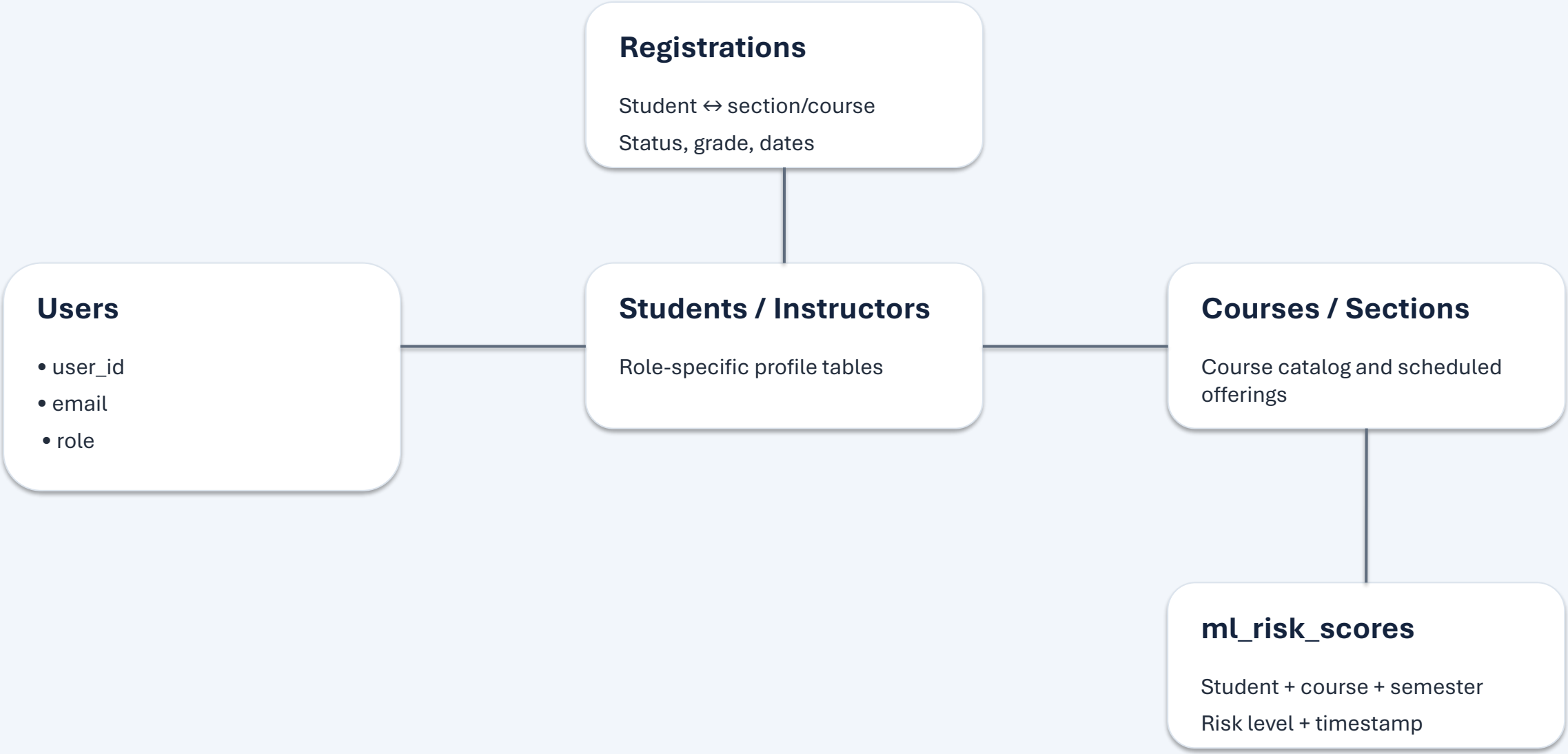
- Users
- Students
- Instructors
- Courses , sections, registrations
- Stored ML risk scores

## Typical request flow

User action → Flask route → parameterized SQL query → MySQL result → Jinja template → rendered dashboard

Predictive analytics currently enters this flow through the ml_risk_scores table.

# Database Design



# System Design — Business Rules & Constraints

### Rule 1: 40-Credit Semester Cap

- **Before adding a course, system calculates:** SUM(registered credits) + new course credits
- If total > 40, registration is blocked
- **Flash message:** 'Cannot register. Maximum 40 credits allowed per semester'

### Rule 2: Graduation Block (140 Credits)

- During login, system checks students.total_credits_completed
- If credits >= 140, session is destroyed immediately
- **Flash message:** 'Your course duration is over (Graduated)'

### Rule 3: Department Filtering & No Duplicates

- Students only see courses from their own department
- Already completed or registered courses are hidden from the catalog

# System Design — ML Risk Prediction Engine

## 5 Input Features

- 1. Student CGPA (0.0 to 10.0)
- 2. Total Credits Completed (academic seniority)
- 3. Course Credit Weight (2 to 8 credits)
- 4. Course Difficulty Index (avg grade of past students)
- 5. Prerequisite Grade (student's grade in prereq course)
- Missing values filled with default mean (7.0)

## How It Works (ml_model.py)

- **Step 1:** Connect to MySQL
- **Step 2:** Read registrations, students, courses, prerequisites
- **Step 3:** Convert grades to numbers (AA=10, FF=0, etc)
- **Step 4:** Calculate course difficulty = avg grade per course
- **Step 5:** Normalize features with StandardScaler
- **Step 6:** Train LogisticRegression (3 classes)
- **Step 7:** Predict risk for all active student-course pairs
- **Step 8:** Save predictions to ml_risk_scores table

## 3 Risk Categories

- **Low Risk (Green badge):** Student likely to score AA/AB/BB
- **Medium Risk (Amber badge):** Student likely to score BC/CC
- **High Risk (Red badge):** Student likely to score CD/DD/FF
- Displayed on student dashboard next to each course
- Helps students make informed decisions before enrolling

# Technology Used

## Database: MySQL 8.0

- Foreign key constraints across all 11 tables
- ENUM types for role and status validation
- Handles 1,282 users and 20,009+ registrations

## Backend: Python 3 + Flask

- Flask web framework for HTTP routing
- Signed session cookies (secret_key )
- Flash messages for user feedback

## ML & Data: scikit-learn + Pandas

- Pandas for data extraction and table merging
- StandardScaler for feature normalization
- LogisticRegression for multi-class prediction

## Frontend: Jinja2 + Vanilla CSS

- Jinja2 server-side templates ({% for %}, {% if %})
- Dynamic rendering of tables, risk badges, flash alerts
- **CSS classes:** .low-risk (green), .medium-risk (amber), .high-risk (red)
- **Confirm dialog on drop action:** onclick="return confirm(...)"

## Security & Connectors

- **mysql-connector-python:** all queries use %s parameterized placeholders
- **Role-based guard on every route:** session.get('role') check

# Live Demo

### Graduated Student Login Block

- **Email:** kamal.iyer.83@iitg.ac.in | Password: hashed_stu_0083
- **Result:** Login denied — flash 'Your course duration is over (Graduated)'

### Active Student Dashboard

- **Email:** tarun.ghosh.244@iitg.ac.in | Password: hashed_stu_0244
- See profile (CGPA 8.10, 80 credits), registered courses, risk badges

### Add Course + 40-Credit Cap Error

- Add courses until total exceeds 40 credits
- System blocks and shows 'Maximum 40 credits allowed' error

### Drop a Course

- Click Drop on a registered course, confirm the prompt
- Course removed from list, credits freed, status = 'Dropped' in DB

### Instructor Dashboard

- **Email:** kapoor.3@iitg.ac.in | Password: hashed_prof_0003
- See assigned sections, room numbers, schedule, student roster with emails

### Admin Dashboard

- **Email:** admin@iitg.ac.in | Password: hashed_admin_01
- See list of all active departments with IDs

# Thank You!